

A decorative graphic on the left side of the slide. It consists of a blue parallelogram and a light green parallelogram, both tilted at an angle. The blue shape is in the foreground, and the green shape is partially behind it. They are set against a dark blue background with subtle diagonal lines.

# Sprint 1 Presentation & Retrospective



# Agenda

Old Business: Team Member Roles & Responsibilities, Project Description, Team Working Agreement, Personas, MVP, Technology Stack.

New Business: Diagrams, Sprint 1 Kanban Status, Metrics.

Demo of Current Product

Retrospective (IdeaBoardz)

Sprint 2 Planning & Goals



# Team

Charles Varga - DevOps/Tech Architect

Tara Edney - Frontend

Andeen Daley (Data Engineer/Database)

Younes Roumila (Backend)

Bella Gerken (AI/NLP Integration)

QA/QC 1 - Charles Varga

QA/QC 2 - James Simpson



# Improvements from Feedback

- Using IdeaBoardz for retrospective



# Project Description

The Intelligent Academic Path Optimizer is a proposed application which will streamline the academic advising workflow. The app is designed to use AI to facilitate decision making when generating a conflict-free academic schedule for students. By utilizing the Constraint Satisfaction Problem algorithms and predictive modeling, the application will dynamically output the most efficient path to graduation based on specific student parameters.



# Team Working Agreement

- Decision Making – Consensus, voting when necessary. Most decisions thus far have been reached by consensus, and the only vote that was taken was for the Project Logo, which was unanimous anyway.
- -Team meetings – Decided using When2Meet on a weekly basis. Team has varying schedules week on week, so this is the best as opposed to a set meeting time.
- -Team keeps team accountable. We will use group accountability and help remind each other of deadlines.
- -Thus far, we have been planning for a few extra days at the end of each sprint just in case things were missed. In Sprint 0, a few items were missed, but we had a few days clear things up.



# Sprint 1 Goals

- Building the core backend for the Python Optimization Engine.
- Drafting the basic visual UI.
- Setting up the initial AI resume parsing pipeline using Gemini.
- “Working Software”
- QA/Testing for User Stories that require them.

# Personas

**Academic Advisor:** Isabelle Johnson, 42 ·  
Pace University

## Problem

- Manually verifies prereqs and builds schedules for 100+ students
- Changing requirements risk graduation delays

## Goals

- Accurate prereq and degree checks
- Automated scheduling
- Earlier risk flags
- More time to help students

## How IAPO Helps

- Generates valid pathways so she can focus on mentorship

**Graduate Student:** William Court, 24 · CS  
Major

## Problem

- Confused about prereqs and availability each semester
- Unclear requirements delay graduation and raise costs

## Goals

- Fastest path to graduate
- "What-if" planning before registering
- Balanced schedule
- Steady GPA

## How IAPO Helps

- Builds an optimized schedule from his credits, goal date, and load

**Dept. Administrator:** Michael Rodriguez, 52 ·  
CS Chair

## Problem

- Little visibility into which prereqs cause delays
- Relies on manual progress reviews

## Goals

- Spot bottlenecks
- Track student progression
- Support data-driven decisions
- Raise grad rates

## How IAPO Helps

- Surfaces bottlenecks from degree plans to guide curriculum planning



# Minimum Viable Product (core features)

- A user-friendly interface in which the user can interact with the AI tied to the app to receive feedback based on the input
- A text input in which students can specify their particular academic needs to the AI
- A clearly visible feedback mechanism which users can utilize to inform developers about the AI, particularly in cases when it generates incorrect information or hallucinates
- A means for users to select the major that they are pursuing and at which level, be it undergraduate or graduate
- Output from the AI which consists of the classes that the students should take based on specified major as well as any text input for specific academic needs in the prompt, and in which semesters these students should take specific classes

# Tech Stack

## **Frontend (React Website)**

- **Figma**
- **React JS**
- **Native CSS**
- **Email JS**

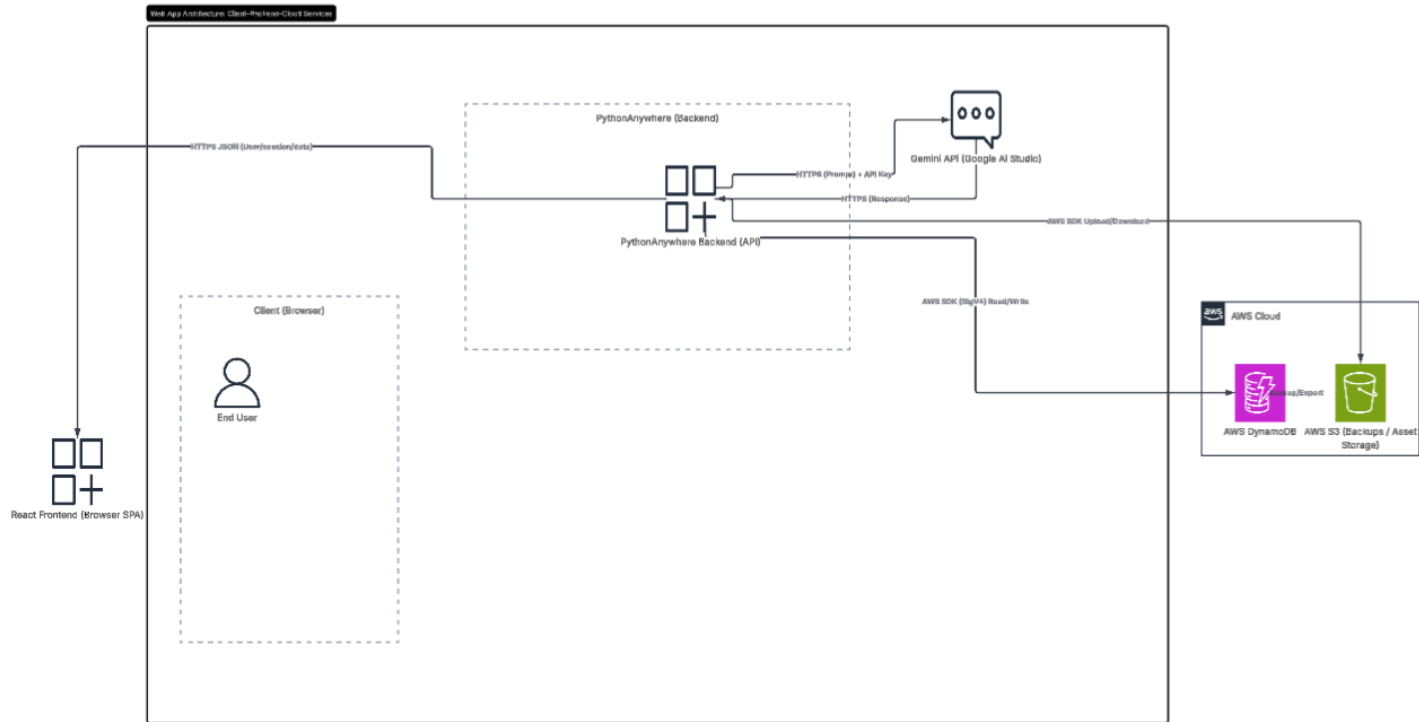
## **Backend (Web Server)**

- **Python 3.12 w.  
FastAPI**
- **Gemini API  
(AI Chat)**
- **AWS  
DynamoDB + S3  
(Accounts)**

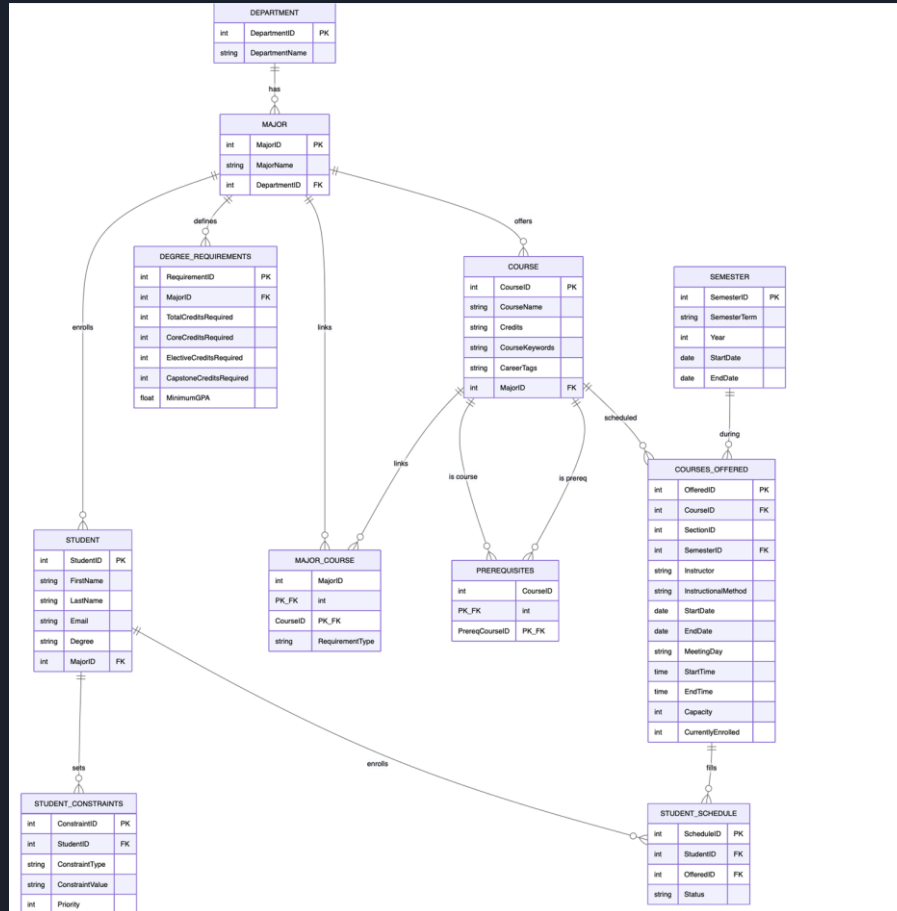
## **Web Hosting**

- **Github Pages  
(Frontend)**
- **PythonAnywhere  
(Web Server)**
- **AWS  
(User Storage)**
- **Gemini (AI Server)**

# Systems Architecture



# Database Design



# Product & Sprint 1 Backlog

<https://github.com/orgs/2026-PaceU-Capstone/projects/2/views/1>

The screenshot displays a GitHub Kanban board for the "IAPO Kanban" project. The board is organized into five columns: Backlog, Ready, In progress, In review, and Done. Each column contains a list of tasks, each represented by a card with a title, story points, and status labels.

**Backlog (9/5, Story Points: 39)**

- intelligent-academic-path-optimizer #26  
Course Availability  
5 Sprint 1
- intelligent-academic-path-optimizer #27  
Prerequisite Validation  
8 Sprint 1
- intelligent-academic-path-optimizer #28  
Student Schedule Review  
5 Sprint 1
- intelligent-academic-path-optimizer #29  
Manage Course Capacity  
5 Sprint 1
- intelligent-academic-path-optimizer #30  
Track Academic Progress  
5 Sprint 1
- Draft  
Add more errors/constraints  
2 Sprint 2 Frontend UI
- Draft

**Ready (9, Story Points: 28)**

- intelligent-academic-path-optimizer #8  
Architect the Optimization Engine Logic  
High 4 Sprint 1 Backend CSP Clear
- intelligent-academic-path-optimizer #25  
Feedback Mechanism  
3 Sprint 1 Clear
- intelligent-academic-path-optimizer #24  
Preference Weighting  
3 Sprint 1 AI Parser Clear
- intelligent-academic-path-optimizer #23  
Transparency / Confidence Flags  
5 Sprint 1 AI Parser Clear
- intelligent-academic-path-optimizer #22  
Transcript Ingestion  
5 Sprint 1 AI Parser Clear
- intelligent-academic-path-optimizer #16  
Check Database interaction with Optimizer  
2 Sprint 1 Database Clear
- intelligent-academic-path-optimizer #15

**In progress (6/3, Story Points: 24)**

- intelligent-academic-path-optimizer #7  
Design the Core Database Schema for Pace MSCS Catalog  
High 4 Sprint 1 Database Clear
- intelligent-academic-path-optimizer #9  
Design the LLM API Integration Pipeline  
High 4 Sprint 1 AI Parser Clear
- intelligent-academic-path-optimizer #10  
Wireframe the Interactive Student Dashboard  
High 4 Sprint 1 Frontend UI Clear
- intelligent-academic-path-optimizer #20  
AI Input Extraction  
8 Sprint 1 AI Parser Clear
- intelligent-academic-path-optimizer #14  
ER diagram  
2 Sprint 1 Database Clear
- intelligent-academic-path-optimizer #31  
Sample Data  
2 Sprint 1 Database Clear

**In review (0/5, Story Points: 0)**

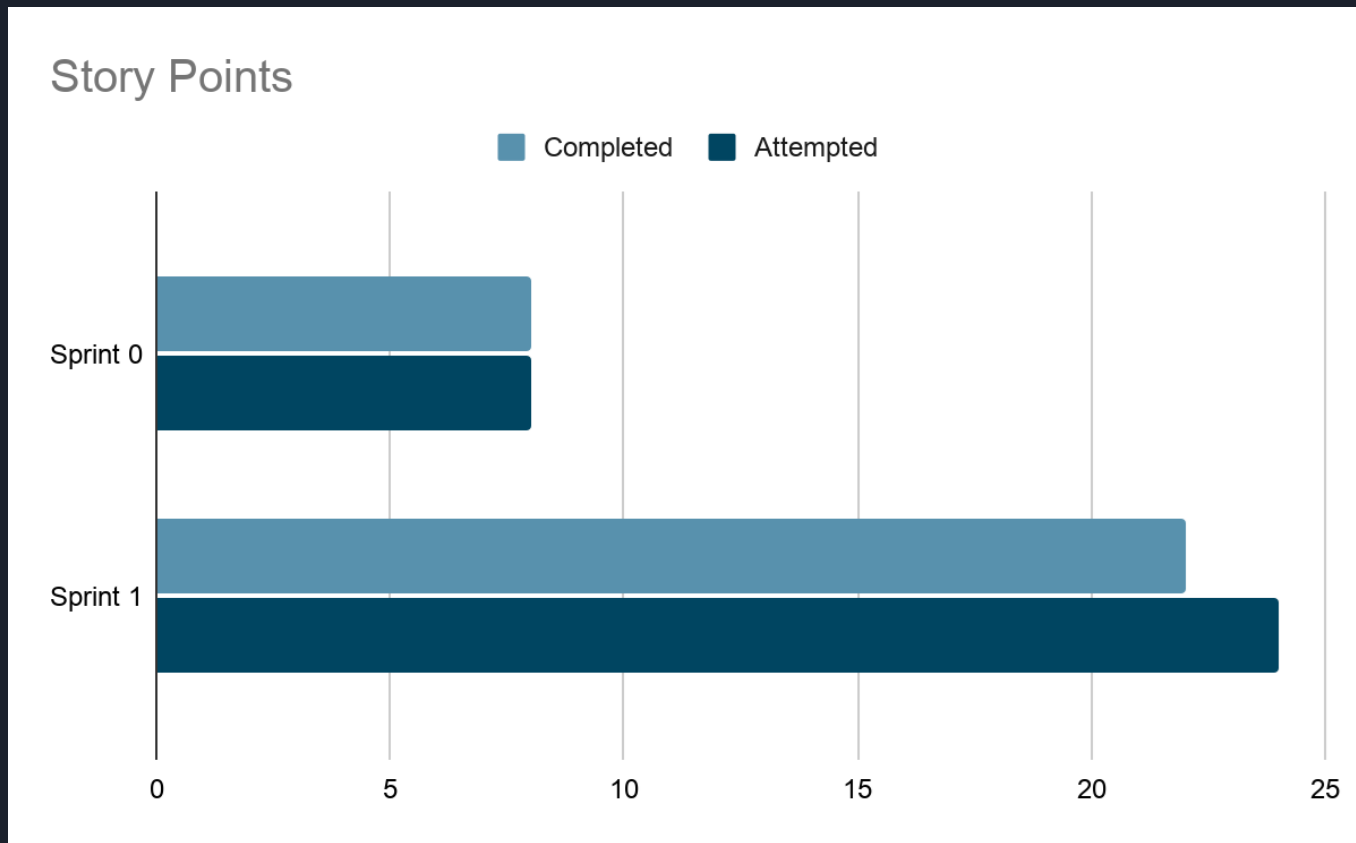
- This item is in review

**Done (4, Story Points: 10)**

- intelligent-academic-path-optimizer #4  
QA/QC of Github Projects Kanban Board  
High 2 Sprint 1 Infrastructure Clear
- intelligent-academic-path-optimizer #11  
Draft the Comprehensive Systems Architecture Diagram  
High 2 Sprint 1 Documentation Clear
- intelligent-academic-path-optimizer #12  
Provision Accounts  
High 4 Sprint 1 Infrastructure Clear
- intelligent-academic-path-optimizer #19  
Data Flow Diagram  
2 Sprint 1 Documentation Clear

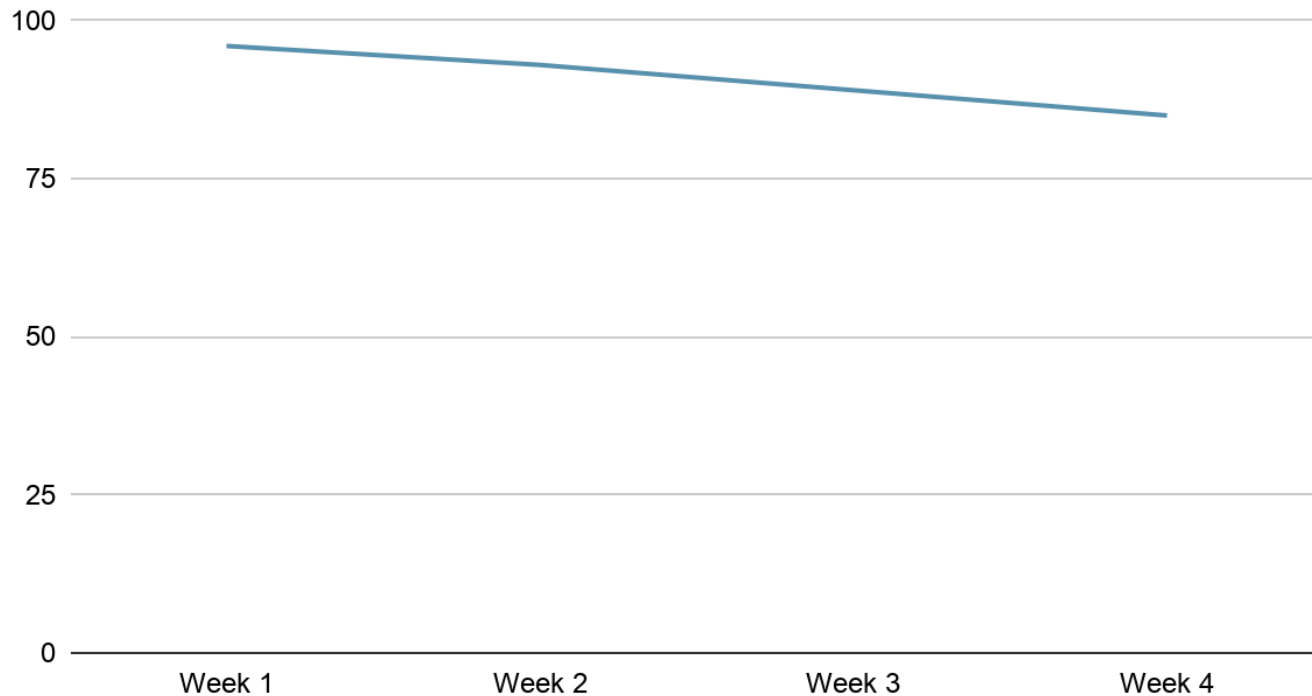
# Metrics

# Team Velocity



# Burndown

## Burndown





# Completed/Committed Ratio

**22/24, 91%**

# Stories Committed to Sprint 2 (All)

<https://github.com/orgs/2026-PaceU-Capstone/projects/2/views/1>

The screenshot displays a GitHub Kanban board for the "2026-PaceU-Capstone" project, specifically the "IAPO Kanban" view. The board is organized into five columns representing different stages of the workflow: Backlog, Ready, In progress, In review, and Done. Each column contains a list of user stories, each with a title, a priority level, and a status (e.g., Sprint 1, High, Backend CSP, Clear). The "Backlog" column shows stories that haven't started. The "Ready" column shows stories ready to be picked up. The "In progress" column shows stories actively being worked on. The "In review" column shows stories currently in review. The "Done" column shows completed stories. The board also includes a search bar at the top, a filter bar, and a "View" button.

**Backlog** 9/5 Story Points: 39  
This item hasn't been started

- intelligent-academic-path-optimizer #26  
Course Availability  
5 Sprint 1
- intelligent-academic-path-optimizer #27  
Prerequisite Validation  
8 Sprint 1
- intelligent-academic-path-optimizer #28  
Student Schedule Review  
5 Sprint 1
- intelligent-academic-path-optimizer #29  
Manage Course Capacity  
5 Sprint 1
- intelligent-academic-path-optimizer #30  
Track Academic Progress  
5 Sprint 1
- Draft  
Add more errors/constraints  
2 Sprint 2 Frontend UI
- Draft

**Ready** 9 Story Points: 28  
This is ready to be picked up

- intelligent-academic-path-optimizer #8  
Architect the Optimization Engine Logic  
High 4 Sprint 1 Backend CSP Clear
- intelligent-academic-path-optimizer #25  
Feedback Mechanism  
3 Sprint 1 Clear
- intelligent-academic-path-optimizer #24  
Preference Weighting  
3 Sprint 1 AI Parser Clear
- intelligent-academic-path-optimizer #23  
Transparency / Confidence Flags  
5 Sprint 1 AI Parser Clear
- intelligent-academic-path-optimizer #22  
Transcript Ingestion  
5 Sprint 1 AI Parser Clear
- intelligent-academic-path-optimizer #16  
Check Database interaction with Optimizer  
2 Sprint 1 Database Clear
- intelligent-academic-path-optimizer #15

**In progress** 6/3 Story Points: 24  
This is actively being worked on

- intelligent-academic-path-optimizer #7  
Design the Core Database Schema for Pace MSCS Catalog  
High 4 Sprint 1 Database Clear
- intelligent-academic-path-optimizer #9  
Design the LLM API Integration Pipeline  
High 4 Sprint 1 AI Parser Clear
- intelligent-academic-path-optimizer #10  
Wireframe the Interactive Student Dashboard  
High 4 Sprint 1 Frontend UI Clear
- intelligent-academic-path-optimizer #20  
AI Input Extraction  
8 Sprint 1 AI Parser Clear
- intelligent-academic-path-optimizer #14  
ER diagram  
2 Sprint 1 Database Clear
- intelligent-academic-path-optimizer #31  
Sample Data  
2 Sprint 1 Database Clear

**In review** 0/5 Story Points: 0  
This item is in review

**Done** 4 Story Points: 10  
This has been completed

- intelligent-academic-path-optimizer #4  
QA/QC of Github Projects Kanban Board  
High 2 Sprint 1 Infrastructure Clear
- intelligent-academic-path-optimizer #11  
Draft the Comprehensive Systems Architecture Diagram  
High 2 Sprint 1 Documentation Clear
- intelligent-academic-path-optimizer #12  
Provision Accounts  
High 4 Sprint 1 Infrastructure Clear
- intelligent-academic-path-optimizer #19  
Data Flow Diagram  
2 Sprint 1 Documentation Clear

# Project Demo

-Frontend Demo

-Backend Demo

-AI Demo

# Frontend Demo

# Backend Demo

AI Demo

# Retrospective

<https://ideaboardz.com/for/Sprint%201%20Retrospective/5796994>

## Sprint 1 Retrospective

### What Went Well +

|                                                                            |                                            |
|----------------------------------------------------------------------------|--------------------------------------------|
| Teamwork<br>+ 4                                                            | Patience and understanding<br>+ 3          |
| Completion of the data flow diagram and the database schema diagram<br>+ 2 | Completion of the Python web server<br>+ 3 |
| Compromising on ideas<br>+ 1                                               |                                            |

### What Can Be Improved +

|                                                                                                                         |                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Maybe more input from people on the updates that get posted for the app? Even if it's just a thumbs up reaction!<br>+ 3 | Communications seem to have slowed down a bit more during this sprint. So that is something that should probably be improved on for the next<br>+ 1 |
| Time management<br>+ 4                                                                                                  | Logging work on Kanban Board<br>+ 2                                                                                                                 |

### Action Items +

|                                                                       |                             |
|-----------------------------------------------------------------------|-----------------------------|
| Increased Communication<br>+ 4                                        | Kanban Board Updates<br>+ 2 |
| Connect the AI Extraction layer to the frontend and the solver<br>+ 2 |                             |

# Links:

Github: <https://github.com/htmw/CS691-Summer2026-Team1/wiki#sprint-0>

Kanban Board: <https://github.com/orgs/2026-PaceU-Capstone/projects/2/views/1>